



KATHMANDU MODEL SECONDARY SCHOOL
First Term Examination- 2080

Class: XI 'Science'

SET-B

Subject: Mathematics [0071]

F.M: 75

P.M: 30

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [11×1=11]

- The AM and HM between two number are 27 and 12 respectively then GM between two numbers be
a) 18 b) 9 c) 36 d) 45
- The sum of infinite geometric series $\frac{1}{2} + \frac{3}{8} + \frac{9}{32} + \dots$ is
a) 1 b) 2 c) 4 d) doesn't exist
- The value of $\lim_{x \rightarrow 2} \frac{\cos x}{x-2}$ is
a) 1 b) -1 c) 0 d) doesn't exist
- For what value of k , the equation $(3k+1)x^2 + 2(k+1)x + k = 0$ may have reciprocal roots?
a) $-\frac{3}{2}$ b) $-\frac{1}{2}$ c) $\frac{1}{2}$ d) 2
- The value of limit $\lim_{x \rightarrow 4} \frac{x^2-2}{x-4}$ is
a) 0 b) $\frac{1}{4}$ c) $\frac{3}{4}$ d) None
- The compound statement $p \wedge (\sim p)$ is
a) tautology b) contradiction c) implication d) inverse
- Both roots of the equation $x^2 + ax + b = 0$ and $x^2 + dx + e = 0$ are common then
a) $ab = de$ b) $d^2e = a^2b$ c) $ae = bd$ d) $ad = be$
- $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$ is
a) 0 b) 1 c) ∞ d) doesn't exist
- Let 'p' be "10 is an even number" and 'q' be "10 is multiple of 5" then symbolic representation of "10 is neither even nor multiple of 5" is
a) $p \wedge q$ b) $(\sim p) \vee (\sim q)$ c) $p \vee q$ d) $(\sim p) \wedge (\sim q)$

- If $x = \{a, b, c, d\}$ then total number of non-empty proper subsets of A is -
a) 4 b) 14 c) 15 d) 16
- If A and B are disjoint sets then which of the following is true?
a) $A \subseteq B$ b) $B \subseteq A$ c) $A \subseteq \bar{B}$ d) $B \subseteq \bar{A}$

GROUP-B [B×5=40]

- Define subset of a set with example.
b) For non-empty sets A, B and C and universal set U, prove that
 $\overline{A \cap B} = \bar{A} \cup \bar{B}$
i) $A - (B \cap C) = (A - B) \cup (A - C)$ [1+2+2]
- a) Define tautology and contradiction of statement with example.
b) For any statement p and q, test whether $(p \wedge q) \wedge (\sim p \vee q)$ is a contradiction or not. [2+3]
- a) If one root of the equation $ax^2 + bx + c = 0$ ($a \neq 0$) be square of the other, then prove that $b^2 + a^2c + ac^2 = 3abc$ [2]
b) Under what conditions the roots of quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$) are
i) real and unequal ii) imaginary iii) equal iv) equal in magnitude but opposite in sign [2]
- For what value of k, the equation $9x^2 + kx + 1 = 0$ has equal roots? [1]
- a) The sum of three numbers in AP is 15, if 1, 4, 19 are added to them respectively, the resulting series is in GP. Find the numbers. [2]
b) If a, b, c are in HP, then prove that $\frac{a+b}{bc}, \frac{b+c}{ca}, \frac{c+a}{ab}$ are in AP. [3]
- a) Define the limit of a function at a point. [1]
b) A function given by, $f(x) = \frac{x^2-16}{x-4}$. Does f(4) exist? Also find $\lim_{x \rightarrow 4} f(x)$ (if exist) [2]
c) Evaluate: $\lim_{x \rightarrow 4} \frac{x^2-2}{x-4}$ [2]
- Evaluate:
a) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$ b) $\lim_{x \rightarrow y} \frac{x \tan x - y \tan y}{x-y}$ [2+3]
- i) What is the value of $\lim_{x \rightarrow 0} \frac{x^n - a^n}{x - a}$ for all rational n and $a \neq 0$ [1]
ii) If $A \subseteq B$ then show that $\bar{B} \subseteq \bar{A}$ [1]
iii) Write the negation of the statement "If 10 is odd or 12 is even, then $5 > 7$ " [1]



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iv) Find the quadratic equation whose one root is $-2 + \sqrt{3}$ [1]

v) Find the value of $3^{1/3}, 3^{1/9}, 3^{1/27}, \dots$ [1]

19. a) Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos x} - 1}{\frac{\pi}{2} - x}$ [2]

b) If the quadratic equations $x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root then prove that the root must be either $\frac{pq' - p'q}{q - q'}$ or $\frac{q - q'}{p' - p}$ [3]

GROUP-B [3×8=24]

20. a) A square is drawn by joining the mid points of the sides of a given square. A third square is drawn inside the second square in the same way and this process continues indefinitely. If a side of the first square is 16 cm, determine the sum of the areas of all the square. [3]

b) The sum of an infinite number of terms of a GP is 23 and sum of their square is 69. Determine the first term and the common ratio. [2]

c) If the roots of the equation $ax^2 + cx + c = 0$ are in the ratio $m:n$, then prove that: $\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + \sqrt{\frac{c}{a}} = 0$ [3]

21. a) Define symmetric difference of two sets A & B. For any sets A and B, prove that $A \Delta B = (A \cup B) - (A \cap B)$ [1+3]

b) Prove that for any two statements p and q

i) $\sim(p \vee q) \equiv \sim p \wedge \sim q$ [2]

ii) $\sim(p \vee (\sim q)) \equiv (\sim p) \wedge q$ [2]

22. Evaluate the following limits: **Fb Page- RS Virtual Classes**
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a) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$ [2]

b) $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x+5} - \sqrt{x})$ [2]

c) $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\cos \theta - \sin \theta}{\theta - \frac{\pi}{4}}$ [2]

d) If $f(x) = \frac{ax+b}{x-1}$, $\lim_{x \rightarrow 0} f(x) = 2$ and $\lim_{x \rightarrow \infty} f(x) = 5$, find the value of $f(4)$ [2]

Best of luck!



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10. The compound statement $p \wedge (\sim p)$ is

- a) tautology b) contradiction c) implication d) inverse

11. The sum of the infinite geometric series $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} \dots$ is

- a) 0.5 b) 1 c) 1.5 d) 3

GROUP-B [8×5=40]

12. a) Define subset of a set with example.

b) For non-empty sets A, B and C and universal set U, prove that

i) $\overline{A \cap B} = \overline{A} \cup \overline{B}$

ii) $A - (B \cap C) = (A - B) \cup (A - C)$

[1+2+2]

13. a) Define tautology and contradiction of statement with example.

b) For any statement p and q, test whether $(p \wedge q) \wedge \sim(p \vee q)$ is a contradiction or not.

[2+3]

14. a) If one root of the equation $ax^2 + bx + c = 0$ ($a \neq 0$) be square of the other, then prove that $b^3 + a^2c + ac^2 = 3abc$

[2]

b) Under what conditions the roots of quadratic equation

[2]

$ax^2 + bx + c = 0$ ($a \neq 0$) are

i) real and unequal

ii) imaginary

iii) equal

iv) equal in magnitude but opposite in sign

c) For what value of k, the equation $9x^2 + kx + 1 = 0$ has equal roots? [1]

15. a) The sum of three numbers in AP is 15. If 1, 4, 19 are added to them respectively, the resulting series is in GP. Find the numbers.

[2]

b) If a, b, c are in HP, then prove that $\frac{b+c}{bc}, \frac{a+c}{ac}, \frac{a+b}{ab}$ are in AP.

[3]

16. a) Define the limit of a function at a point.

[1]

b) A function given by; $f(x) = \frac{x^2-16}{x-4}$. Does f(4) exist? Also find

[2]

$\lim_{x \rightarrow 4} f(x)$ (if exist)

c) Evaluate: $\lim_{x \rightarrow 64} \frac{\sqrt[6]{x}-2}{\sqrt[3]{x}-4}$

[2]

17. Evaluate:

a) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$

b) $\lim_{x \rightarrow y} \frac{x \tan x - y \tan y}{x - y}$

[2+3]

18. i) What is the value of $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$ for all rational n and $n \neq 0$

[1]

ii) If $A \subseteq B$ then show that $\overline{B} \subseteq \overline{A}$

[1]

iii) Write the negation of the statement "If 10 is odd or 12 is even, then $5 > 7$ "

[1]

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KATHMANDU MODEL SECONDARY SCHOOL

First Term Examination- 2080

Class: XI 'Science'

SET-A

F.M: 75

Time: 3 hrs.

Subject: Mathematics [0071]

P.M: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP-A [MCQ] [11×1=11]

- For what value of k , the equation $(3k+1)x^2 + 2(k+1)x + k = 0$ may have reciprocal roots?
 - $-\frac{3}{2}$
 - $-\frac{1}{2}$
 - $\frac{3}{2}$
 - 2
- If $A = \{a, b, c, d\}$ then total number of non-empty proper subsets of A is ...
 - 4
 - 14
 - 15
 - 16
- Both roots of the equation $x^2 + ax + b = 0$ and $x^2 + dx + e = 0$ are common then
 - $ab = de$
 - $d^2e = a^2b$
 - $ae = bd$
 - $ad = be$
- A and B are two non-empty sets then which one is correct?
 - $(A \cup B) \subset (A \cap B)$
 - $(A \cap B) \subset (A \cup B)$
 - $(A \cup B) = (A \cap B)$
 - $(A \cup B) - (A \cap B) = \emptyset$
- Let 'p' be "10 is an even number" and 'q' be "10 is multiple of 5" then symbolic representation of "10 is neither even nor multiple of 5" is
 - $p \wedge q$
 - $(\sim p) \vee (\sim q)$
 - $p \vee q$
 - $(\sim p) \wedge (\sim q)$
- The value of $\lim_{x \rightarrow \infty} x \cdot \sin \frac{1}{x}$
 - 1
 - 1
 - 0
 - doesn't exist
- $\lim_{x \rightarrow 0} \frac{\ln(x+1)}{e^x - 1} = ?$
 - 1
 - 0
 - 1
 - doesn't exist
- The AM and HM between two number are 27 and 12 respectively then GM between two numbers be
 - 18
 - 9
 - 36
 - 45
- The value of limit $\lim_{x \rightarrow 4} \frac{x^2 - 2x - 8}{x - 4}$ is
 - 0
 - 2
 - 8
 - 6

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Class: XI 'Science'

First-Term Examination 2080

Time: 3hrs.

SET - B

F.M.: 75

Subject: Physics '1011'

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'A' [11×1=11]

Multiple choice questions

- 1) If pressure P , velocity V and time T are taken as fundamental physical quantities, the dimensional formula of force is
 - a) $[P^{-1}VT^2]$
 - b) $[PV^2T^2]$
 - c) $[P^{-1}V^{-1}T^{-2}]$
 - d) $[PVT^2]$
- 2) If the magnitude of difference of two-unit vectors is $\sqrt{3}$, then the magnitude of sum of the two vectors is
 - a) 1
 - b) 2
 - c) 3
 - d) $\sqrt{3}$
- 3) Which of the following cannot be resultant of the vectors of the magnitude 8 and 4?
 - a) 2
 - b) 4
 - c) 6
 - d) 12
- 4) A force of 20 N is applied at an angle of 60° to the vertical. What is the horizontal component of this force?
 - a) 20N
 - b) 10N
 - c) 17.32N
 - d) 15N
- 5) The maximum and minimum magnitude of the resultant of two given vectors is 17 units and 7 units respectively. If these two vectors are at right angles to each other, the magnitude of resultant is
 - a) 14
 - b) 16
 - c) 18
 - d) 18
- 6) Y-component of velocity is 20 and x-component is 10. The direction of motion of the body with the horizontal at this instant is
 - a) $\tan^{-1} \frac{1}{2}$
 - b) 45°
 - c) $\tan^{-1} 2$
 - d) 0°
- 7) An ant moves towards the plane mirror with speed of 2m/s and the mirror is moved towards the ant with the same speed. What is the velocity of ant with respect to its image?
 - a) 2 m/s
 - b) 4 m/s
 - c) 8 m/s
 - d) 16 m/s
- 8) The focal length of concave mirror is f and the distance from the object to the pole is x . The magnification is
 - a) $\frac{f}{f-x}$
 - b) $\frac{x}{f}$
 - c) $\frac{f}{x}$
 - d) $\frac{f}{f+x}$
- 9) A point object is placed on the axis of the concave mirror at a distance of 45 cm from the focal point of the mirror. Its image is formed at the point of object then focal length of the mirror is
 - a) 90 cm
 - b) 135 cm
 - c) 45 cm
 - d) 15 cm
- 10) Two equally charged, identical metal spheres A and B repel each other with force F . The spheres are fixed with a distance r between them. Third identical but uncharged sphere is brought in contact with A and then placed at midpoint of line joining A and B. The magnitude of net electrostatic force on third sphere is
 - a) F
 - b) $2F$
 - c) $3F/4$
 - d) $F/2$
- 11) A faulty thermometer with lower fixed point and upper fixed point 5°C and 95°C respectively. What is the correct temperature when the reading gives 35?
 - a) 60°C
 - b) 40°C
 - c) 33.3°C
 - d) 20.5°C

GROUP 'B' [8×5=40]

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Short answer questions:

- 12) a) What are the limitations of dimensional analysis?
- b) Check the correctness of the physical relation $\rho = \frac{3g}{4\pi r G}$ where symbols carry their usual meanings.
- c) Find the value of 20 J on the system based on 20 cm, 1kg and $\frac{1}{2}$ minute as fundamental units.